Error Decomposition in Training-Free Single-FLAIR Lower-Grade Glioma Segmentation: Generation versus Label-Free Selection

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Highlights

- Error decomposition separates localization, representation, and selection.
- Expanded candidates raise oracle Dice from 0.872 to 0.906.
- The canonical pool covers 103/110 cases at Dice$\geq$ 0.70.
- Weak boundaries predict lower candidate ceilings across three cohorts.
- Frozen single-FLAIR segmentation achieves 0.712 Dice on UCSF-PDGM.

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Abstract

Training-free single-FLAIR glioma segmentation may fail through unsuccessful localization, inadequate boundary representation, or selection of an inferior candidate. We separate these error sources using a fixed method without learned weights that combines edge-preserving direction-dependent diffusion, score-topology threshold sweeping, and closed-form label-free selection. Fuzzy α-cut candidates are added only for retrospective capacity analysis. The whole-tumour target is evaluated on the ground-truth-selected largest-tumour axial slice from TCGA-LGG development data (N=110), an independent UCSF-PDGM WHO grade 2–3 cohort (N=99), and BraTS 2023 (N=150) as a cross-grade stress test. Ground truth supported slice selection, fixed-parameter development on TCGA-LGG, and retrospective evaluation, but not case-level candidate generation or selection after fixation. On TCGA-LGG, mean delivered Dice is 0.789 and the candidate-pool ceiling is 0.872. Localization succeeds in 106/110 cases; 103/110 contain a candidate with Dice$\geq$ 0.70, while 16/103 representable cases retain a selection gap of at least 0.10. Nested seed and α-cut expansion raises the retrospective ceiling to 0.906 and satisfies the prespecified 0.005 practical-saturation margin, without changing delivered Dice. Weak-boundary fraction is negatively associated with the pool ceiling in TCGA-LGG ($\rho = -0.537$), UCSF-PDGM ($\rho = -0.438$), and BraTS 2023 ($\rho = -0.467$), with direction preserved after covariate adjustment. The frozen method achieves 0.712 mean Dice on UCSF-PDGM without retuning. Thus, boundary-dependent candidate generation and label-free selection constitute distinct bottlenecks that cannot be identified from the delivered mask alone.

Keywords: lower-grade glioma; FLAIR MRI; training-free segmentation; candidate generation; label-free selection; error decomposition; candidate-pool ceiling

1 Introduction

Glioma segmentation supports quantitative assessment of tumour burden and longitudinal change. Contemporary performance is dominated by supervised networks trained with expert masks and multiple co-registered MR sequences [1, 2]. Those conditions are not universal: retrospective archives may contain only one sequence, and expert delineations are costly. Fluid-attenuated inversion recovery (FLAIR) is particularly relevant in this restricted setting because it suppresses cerebrospinal fluid and highlights much of the whole-tumour abnormality. The scientific question addressed here is therefore not whether a handcrafted method can replace multi-modal supervised systems, but what can be achieved when only FLAIR is available and neither training nor learned weights are permitted.

Lower-grade glioma is a difficult instance of this problem. Its appearance varies substantially across patients, while infiltrative margins, partial-volume effects, and intensity inhomogeneity weaken both intensity and edge assumptions [3, 4]. A lesion may be almost isointense with surrounding parenchyma, and its boundary may be less conspicuous than unrelated anatomical interfaces. Consequently, a poor final mask does not identify a unique failure mechanism. The method may never reach the tumour; it may reach the tumour but fail to generate its boundary; or it may generate an adequate mask and then select a distractor. These failures require different remedies, yet a single delivered Dice score collapses them into one number.

We formulate this ambiguity as a three-stage error decomposition. *Localization* asks whether any candidate reaches a sufficient fraction of the tumour. *Boundary representation* asks whether the candidate family contains a mask with adequate overlap. *Selection* asks whether a closed-form, label-free criterion chooses such a mask when it is available. The best candidate identified retrospectively with ground truth is termed the *pool oracle*. It is a diagnostic measure of candidate-family capacity, not an inference output or a claim about the ceiling of all training-free methods. Joint reporting of the delivered mask and the pool oracle separates failure to generate from failure to select.

The primary method implements a generate–select strategy. A NewMetric direction-dependent diffusion front end produces an edge-preserving field from FLAIR [5]. Its delivered candidates are generated by seeded superlevel-set sweeping of a score field, and a label-free criterion ranks them using region quality and threshold persistence. Direct fuzzy α-cuts of intensity membership are introduced separately in the retrospective candidate-capacity analysis. Two intensity windows are evaluated within one common pool to accommodate bright-but-not-extreme lower-grade lesions without making tumour grade an inference-time input. NewMetric has a traceable Beltrami–Beil–Finsler derivation, but the subsequent traversal is performed on Euclidean image-grid superlevel components; the method is therefore not presented as Finsler-geodesic segmentation. Nor is improved accuracy uniquely attributed to the direction-dependent front end.

The evaluation deliberately separates inference from measurement. No labels or learned parameters enter candidate generation or selection. Ground truth is used to choose the largest-tumour axial slice and, retrospectively, to compute performance, pool ceilings, and operational diagnostic states. The experiment consequently evaluates fixed-slice segmentation rather than automatic slice detection or 3D tumour localization. TCGA-LGG ($N = 110$) is the development and decomposition cohort; UCSF-PDGM WHO grade 2–3 ($N = 99$) provides frozen-parameter external validation; and BraTS 2023 ($N = 150$) is used only as a cross-grade stress test.

The contributions are fourfold. First, the study provides an operational localization–boundary–selection decomposition and reports it alongside label-free delivery. Second, it tests whether weak tumour boundaries are associated with candidate-family capacity across three cohorts. Third, nested seed and α-cut budgets quantify practical saturation of the candidate ceiling separately from selection accuracy. Fourth, matched candidate-channel, selector-robustness, and unsupervised-baseline controls define which gains belong to generation and which remain inaccessible to the label-free selector. The result is an evaluation framework for locating error in training-free candidate systems, rather than a claim of parity with supervised multi-modal segmentation.

2 Related Work and Study Positioning

Brain tumour segmentation studies occupy different supervision and input regimes, and their reported Dice values are not interchangeable. Supervised U-Net-type systems use pixel-wise labels and commonly combine T1, post-contrast T1, T2, and FLAIR [1, 2, 6]. Supervised single-FLAIR systems have also been evaluated on TCGA-LGG and related FLAIR data [7, 8], but their learned spatial features and labelled training protocol differ fundamentally from the present setting. At another point on the spectrum, SAM and MedSAM generate masks from box or point prompts [9, 10]. If such prompts are derived from the reference mask, the resulting score includes oracle localization; if prompts are generated automatically, prompt discovery becomes a separate localization task. These systems are therefore important performance references, but not matched comparators for prompt-free, training-free inference.

This distinction also motivates the term *training-free*. In this paper it means that inference uses neither a fitted predictive model nor learned weights; the handcrafted constants were nevertheless developed with reference-mask performance available on TCGA-LGG and then frozen. It is narrower than the often-used label-free or unsupervised designation, which may include representation learning or healthy-cohort anomaly models. Comparisons are consequently restricted by four protocol axes: supervision, modality, dimensionality, and target definition. The present protocol is training-free, single-FLAIR, two-dimensional, and targets the whole tumour on a ground-truth-selected slice.

Within the training-free regime, classical segmentation includes thresholding, clustering, seeded region growing, watershed methods, and active contours [3, 11]. These methods remain relevant because their decisions are traceable, but they depend strongly on intensity separability, initialization, and stopping rules. Multilevel Otsu- or entropy-based thresholding with metaheuristic search exemplifies single-mask intensity optimization [12]. Fuzzy clustering, originating from the fuzzy c-means formulation [13], and level-set combinations instead construct an initial region and evolve its boundary. Khosravanian et al. [14] reported high Dice on 40 selected BraTS 2017 FLAIR slices, whereas Darwish et al. [15] reported a high volumetric Dice on BraTS 2019. Those results were obtained under different sample-selection, dimensionality, and optimization protocols and are cited as method-family references rather than head-to-head performance estimates.

AUCseg is a particularly relevant unsupervised reference because its original three-dimensional, multiparametric framework first extracts whole tumour from T2-FLAIR by clustering before segmenting subregions with additional modalities [17]. The matched control used here reimplements only that whole-tumour concept under the present two-dimensional, single-FLAIR protocol; it is not a reproduction of the complete AUCseg system.

Candidate-pool methods expose an additional problem that single-mask methods hide. Biratu et al. [16] generated multiple automatically seeded region-growing candidates, but reported the best candidate selected against ground truth. This distinction is also consistent with recent work on ground-truth-free segmentation evaluation [18]. The best-candidate quantity is directly analogous to a pool oracle: it measures whether the generator can produce a suitable region, not whether a deployable label-free selector can identify it. The distinction is central here. We report both the label-free delivery and the retrospective pool oracle and retain zero-Dice cases in cohort means.

The primary and capacity-analysis candidate channels address complementary limitations. Seeded score-topology sweeping preserves connectivity to multiple plausible locations but can miss a lesion when no seed reaches it or when a low-contrast bridge causes leakage. Fuzzy α-cuts provide seed-independent, nested intensity-membership boundaries, but do not encode the plateau structure of the score field. Their union broadens representational capacity; it does not, by itself, solve candidate selection.

The connected components of all superlevel sets of a scalar image form a max-tree under set inclusion, a standard morphological hierarchy for connected filtering [22, 23]. Components containing one fixed point form a totally ordered chain within that tree [24]; following this chain is therefore a well-defined seed-conditioned traversal rather than an arbitrary choice at component merges. Stability under threshold change is also the organizing principle of maximally stable extremal regions [25]. The present method uses these ideas in a restricted form: it samples the seed-containing superlevel component at 200 prescribed thresholds and records runs with limited area growth. The resulting π is a sampled plateau length, not classical birth–death persistence [19], a complete max-tree attribute, or a higher-dimensional topological prior [21]. A long plateau indicates threshold stability but cannot establish that the component is tumour; tumour specificity must come from the remaining image evidence and be tested empirically.

The literature commonly emphasizes candidate construction or final overlap, whereas fewer studies quantify whether a good candidate existed before it was selected. This omission matters because improving the generator can increase the number of plausible distractors and enlarge the selection problem. Conversely, a better selector cannot recover a boundary absent from the pool. The present study therefore treats pool capacity and label-free delivery as separate endpoints and uses nested candidate budgets to determine whether the observed ceiling reflects an undersampled family or a practical saturation of that family.

Taken together, the closest methodological comparators are not defined by their reported Dice alone, but by whether they are training-free, single-modality, prompt-free, and candidate-based. Within this space, the contribution is an error-accounting framework: candidate-reach limitation, boundary-representation limitation, and selection loss are assigned study-specific operational definitions and measured separately. The NewMetric front end provides a geometrically derived edge-preserving field, but the principal claim does not require Finsler-specific accuracy superiority. The novel empirical question is instead whether a candidate family can represent the tumour boundary and, conditional on that representation, whether an image-derived criterion can select it without ground truth. This positioning prevents pool oracles, selected-subset results, prompted outputs, and supervised models from being interpreted as equivalent forms of automatic training-free performance.

3 Materials and Methods

3.1 Study Design and Experimental Protocol

This study evaluates training-free segmentation of a prespecified axial FLAIR slice. The method receives only the FLAIR image, requires no learned parameters, and does not access the reference mask during candidate generation or label-free selection. Ground truth is used for five explicitly separated experimental purposes: selection of the largest-tumour slice, development of fixed parameters on TCGA-LGG, retrospective performance evaluation, diagnostic error decomposition, and computation of candidate-family ceilings. Consequently, the experiment addresses segmentation conditional on a tumour-bearing slice; it does not constitute end-to-end tumour detection or volumetric segmentation.

Throughout the study, a binary prediction is denoted by A, its nonempty binary whole-tumour reference mask by Y, and their Dice coefficient by

$$D(A,Y) = \frac{2|A \cap Y|}{|A| + |Y|} \in [0,1], \tag{1}$$

where $|\cdot|$ denotes pixel cardinality; $D(A, Y) = 0$ when A is empty.

The cohorts have distinct and predefined roles. TCGA-LGG serves as the development and internal evaluation cohort. All fixed parameters are selected on this cohort with reference-mask performance available and are frozen before external evaluation. The WHO grade 2–3 subset of UCSF-PDGM serves as the independent lower-grade external-validation cohort. BraTS 2023 GLI serves as a cross-grade and preprocessing-domain stress test rather than as lower-grade validation. No parameter is re-estimated on either external cohort.

Two analyses must be distinguished. The *primary label-free method* denotes the fixed candidate-generation and selection procedure that reports delivered segmentation performance. The *expanded candidate-capacity analysis* enlarges the seed and fuzzy α-cut budgets in nested fashion, where $\alpha \in [0, 1]$ is a fuzzy-membership threshold, and uses ground truth only to measure the best attainable candidate in each set. Its pool-oracle Dice is a retrospective upper bound on boundary availability, not a segmentation result available during label-free application. This distinction is maintained throughout the method, evaluation and results.

3.2 Imaging Cohorts and Reference Masks

TCGA-LGG [7] — *the focus set of the study*. Lower-grade (WHO II–III) gliomas, described using the contemporaneous cohort terminology [26], with low-to-moderate contrast and comparatively regular boundaries. The local analysis copy was derived from the public 256×256 three-channel TCGA-LGG derivative by extracting channel 1, the FLAIR channel. This extraction was verified against all source TIFF files; the pre-contrast and post-contrast channels were neither combined with nor supplied to the method. All tumour-bearing 2D axial slices were extracted (1360 *slices in total*); as required by the evaluation protocol, exactly **one** slice per patient — the one with the largest ground-truth tumour area — was retained, yielding an experimental set of 110 slices. These images are therefore described as a *skull-intact, preprocessed 2D derivative*, not as unprocessed DICOM. No additional skull-stripping or modality fusion was applied; a head-support mask was obtained at a normalized-intensity threshold of 0.05, as formally defined in Section 3.3. All parameters were determined on this set alone. Table 1 summarizes the basic properties of the dataset.

Table 1: Descriptive statistics of the TCGA-LGG dataset (skull-intact preprocessed derivative, one largest-tumour slice per patient; 110 cases).

Property	Value
Original source	TCIA — TCGA-LGG collection [7]
Tumour type	Lower-grade glioma (WHO II–III)
Imaging sequence	FLAIR (single modality)
Preprocessing	Skull-intact preprocessed 2D derivative; study-specific scaling to $[0, 1]$
Number of patients	110
Tumour-bearing 2D slices per volume (all)	1 360 (min 3, max 36, mean 12.4)
Slice resolution	256×256 pixels
Head-mask area (normalized-intensity threshold 0.05)	~46–65% of the image area
Ground-truth (GT) tumour area — all 1360 *slices*	
Minimum	50 pixels
Maximum	7 929 pixels
Mean ± SD	1 950 ± 1 525 pixels
Median	1 541 pixels
Ground-truth (GT) tumour area — selected 110 *slices*	
Minimum	582 pixels
Maximum	7 929 pixels
Mean ± SD	3 013 ± 1 770 pixels
Median	2 642 pixels

BraTS 2023 (GLI) [27, 4, 28, 29] — *cross-grade stress test (no tuning).* This cohort, of which only the FLAIR channel is used, consists predominantly of high-grade, skull-stripped and template-aligned cases. The same frozen settings are applied to the ground-truth-selected largest-tumour slices of 150 imaging studies from 143 patients. This cohort is not interpreted as lower-grade external validation; it probes operability under a different tumour grade and preprocessing regime.

UCSF-PDGM [30, 31] — *independent lower-grade external validation (no tuning).* UCSF-PDGM contains WHO grade 2–4 diffuse gliomas imaged with standardized 3T preoperative MRI together with expert-corrected multi-compartment tumour masks. The evaluated subset comprises WHO grade 2 (n=56) and grade 3 (n=43) cases (N=99), using only their FLAIR channel. Known follow-up/recurrence identifiers are grade 4 and do not enter this subset; the grade 2–3 identifiers are unique and none of them carries a BraTS21 identifier. For each case the axial slice with the largest ground-truth whole-tumour area is selected, and all parameters frozen on TCGA are applied unchanged. Because the UCSF images are skull-stripped, this cohort probes the cross-institutional lower-grade generalization of the primary candidate-generation and selection method; it does not independently validate the raw-image ROI correction.

Slice selection and the rationale for the 2D protocol. In all cohorts the unit is *a single axial slice per case*: among the slices of the volume, the one whose binary tumour mask has the largest pixel area is selected. This selection relies directly on the ground truth; the experiment is therefore not an end-to-end label-independent tumour detection protocol, and the reported values do not include automatic volume or slice finding performance. The ground truth is binarized in "whole-tumour" form by taking the union of all labels greater than zero.

There are three reasons for evaluating at the 2D slice level in this study: (i) the focus TCGA-LGG data are organized as per-patient selected 2D FLAIR slices; (ii) for protocol consistency, the same ground-truth largest-tumour-slice rule is applied to UCSF-PDGM and BraTS 2023; and (iii) 3D volumetric extension and automatic slice finding are outside the scope of this study. This protocol compares fixed-slice segmentation; it does not evaluate volumetric tumour detection.

3.3 Image Preparation and Score-Field Construction

Let $\Omega \subset \mathbb{Z}^2$ denote the 256×256 pixel grid, $I_{\mathrm{raw}} : \Omega \to \mathbb{R}_{\geq 0}$ the prespecified FLAIR slice, and $|A|$ the number of pixels in a set $A \subseteq \Omega$. Intensities are scaled to the unit interval, a non-background support is defined, and values within that support are renormalized:

$$I = \frac{I_{\mathrm{raw}}}{\max I_{\mathrm{raw}}}, \qquad B = \{x \in \Omega : I(x) > 0.05\}, \qquad I_n = \frac{I}{\max_B I}. \tag{2}$$

Here $I : \Omega \to [0, 1]$ is the globally normalized image, $B \subseteq \Omega$ is the non-background head support, $I_n : \Omega \to [0, 1]$ is the image renormalized by the maximum within B, and $\max_B$ and $\min_B$ denote extrema over pixels in B. The NewMetric edge-preserving diffusion [5], which belongs to the broader family of edge-preserving anisotropic diffusion methods [32, 33], is applied with $\beta_{\mathrm{NM}} = 5.0$, $\Delta t = 0.15$, and three iterations, where β_{NM} controls the strength of the direction-dependent metric deformation and Δt is the explicit diffusion time step. Its result is rescaled within B to obtain

$$F_n = \frac{\mathrm{NewMetric}(I) - \min_B \mathrm{NewMetric}(I)}{\max_B \mathrm{NewMetric}(I) - \min_B \mathrm{NewMetric}(I)}. \tag{3}$$

$F_n : \Omega \to [0, 1]$ is therefore the normalized edge-preserving NewMetric field used by all subsequent score calculations. All included tumour-bearing images have positive denominators in Eqs. (2)–(3); the implementation nevertheless adds 10^{-8} as a floating-point safeguard, without changing the analytical definitions. Gaussian filtering is not used in the segmentation signal path.

The Sobel gradient magnitude $H = \|\nabla F_n\|_2$ is converted to an edge permeability field using the intra-support 95th percentile $K = P_{95}(H; B)$, where $P_q(Z; A)$ denotes the qth percentile of field Z over pixels in A:

$$g(x) = \frac{1}{1 + [H(x)/K]^2}, \qquad x \in \Omega. \tag{4}$$

Thus $g : \Omega \to (0, 1]$ approaches one in locally homogeneous regions and decreases as the edge magnitude increases relative to K. For each of two complementary intensity windows, a soft band-pass weight and a score field are defined as

$$w(I_n; \tau_h) = \frac{1}{1 + \exp[-(I_n - 0.28)/0.05]} \frac{1}{1 + \exp[(I_n - \tau_h)/0.05]}, \tag{5}$$

$$S(x; \tau_h) = g(x) F_n(x) \mathbf{1}_B(x) w(I_n(x); \tau_h), \tag{6}$$

where $\mathbf{1}_B$ is the indicator of B; 0.28 is the lower intensity transition; 0.05 is the sigmoid softness; and $(\tau_h, p_h) \in \{(0.82, 88), (0.97, 98)\}$. Here τ_h is the upper soft-window transition and p_h is the corresponding upper FLAIR percentile used for seed eligibility. Hence $w \in (0, 1)$ is the intensity-band weight and S is the window-specific score field. To avoid favouring the window from which a candidate originates, final candidate assessment uses the window-independent evidence field $E(x) = g(x) F_n(x) \mathbf{1}_B(x)$.

3.4 Primary Candidate Generation and Label-Free Selection

Seed localization and region evolution deliberately rely on different evidence. Seed locations are local maxima of S restricted to $P_{30}(I_n; B) \leq I_n(x) \leq P_{p_h}(I_n; B)$. At most ten seeds are retained per window, with a minimum separation of 12 pixels.

Seed-conditioned superlevel traversal. For a threshold t, define the score superlevel set $X_t = \{x \in B : S(x) \geq t\}$. If $t_1 > t_2$, then $X_{t_1} \subseteq X_{t_2}$; hence the 4-connected components across decreasing thresholds form an inclusion hierarchy, or max-tree [22, 23]. For a fixed seed s, the components that contain s constitute a nested chain [24]. The traversal used here materializes only this seed-conditioned chain; it neither constructs the full max-tree nor propagates a front by Euclidean or geodesic distance.

For each seed, the method samples

$$t_j = S(s)\left[1 - 0.8\frac{j-1}{199}\right], \qquad j = 1, \ldots, 200, \tag{7}$$

and follows the component

$$C_j(s) = \mathrm{CC}_s(X_{t_j}) = \mathrm{CC}_s\left(\{x \in B : S(x) \geq t_j\}\right). \tag{8}$$

Thus, $t_1 = S(s)$ and $t_{200} = 0.20S(s)$; the 200 thresholds are equally spaced on this seed-specific interval. The operator $\mathrm{CC}_s(A)$ returns the 4-connected component of binary set A containing s, and $C_j(s)$ is its mask at level t_j. Four-connectivity matches the extremal-region convention used by the implementation [25]; eight-connectivity is not evaluated as a separate experimental arm. In the two-dimensional grid, this choice admits horizontal and vertical adjacency but does not merge regions that meet only at a corner. It therefore provides a conservative branch topology: isolated diagonal contacts do not cause an early component merger, do not create a one-pixel diagonal leakage route, and do not introduce the associated jump into the seed-component area trajectory. The trade-off is that a genuinely thin or oblique structure connected only through diagonal pixels can be split. Thus, four-connectivity contributes a deterministic and comparatively conservative topological convention, not a demonstrated accuracy advantage; the latter would require a prespecified four-versus-eight-connectivity ablation. Because superlevel sets are nested, $C_j(s) \subseteq C_{j+1}(s)$ until the traversal terminates.

Leakage-control stops. Let $a_j = |C_j(s)|$ and $\Delta_j = a_j - a_{j-1}$ denote component area and its increment at successive levels, and let $n_{\min} = \max(40, 0.02|B|)$ pixels denote the area at which the jump and acceleration stops become active. Evolution terminates when either $a_j > 0.25|B|$ (the area-cap stop) or the mean of g over the current component boundary is below 0.08 (the edge-wall stop). Once $a_{j-1} \geq n_{\min}$, two additional growth-instability stops are enabled: $a_j > 4a_{j-1}$ (the jump stop), or $\Delta_j > 2.5\Delta_{j-1}$ and $\Delta_{j-1} > 2.5\Delta_{j-2} + 1$, with all three increments positive (the acceleration stop). These are readout safeguards applied to the sampled branch: they do not alter the superlevel connectivity or decide which components merge. The area cap limits gross support capture, the jump and acceleration criteria detect abrupt merging or leakage after a stable-size component has formed, and the edge-wall condition stops a branch whose current boundary lies predominantly on low-permeability pixels.

Plateau extraction. A stable plateau is a maximal run of sampled thresholds $t_r, t_{r+1}, \ldots, t_{r+m-1}$ for which $a_j \leq 1.06a_r$; here a_r is the component area at the first, highest-threshold sample of the run and $m \geq 1$ is its number of sampled threshold levels. This relative-area criterion follows the extremal-region principle that a useful component changes little over a threshold neighborhood [25], while using the one-sided growth trajectory of the fixed seed. Plateau tracking begins once $a_j \geq 30$ pixels. Let $M_{\mathrm{raw}} = C_{r+m-1}(s)$ be the largest component reached within the run. Its holes are filled to obtain M only when the filled area is no greater than $2|M_{\mathrm{raw}}|$; otherwise $M = M_{\mathrm{raw}}$. The persistence attribute is the dimensionless run length $\pi(M) = m \in \{1, \ldots, 200\}$, not the numerical width of the score-threshold interval and not the classical lifetime between a component's birth and death. The seed need not coincide with the local maximum at which its max-tree branch is born, and the sweep begins at $S(s)$ rather than at that birth level; classical zero-dimensional persistence terminology would therefore overstate what is measured [19]. Seeds are determined from a FLAIR-restricted score maximum, whereas candidate boundaries are determined by this sampled superlevel-component traversal.

For every nonempty candidate mask $M \subseteq B$, let $p(M)$ be the sum of the piecewise-Euclidean lengths of its external OpenCV contours, and let $h(M)$ be the sum of the continuous areas of their convex hulls. Define compactness as $c(M) = \mathrm{clip}(4\pi|M|/p(M)^2, 0.01, 1)$ and solidity as $\sigma(M) = |M|/h(M)$ when $h(M) > 0$ (and zero otherwise). Label-free quality then combines area, mean evidence, compactness and solidity:

$$Q(M) = |M|\, \overline{E}_M\, c(M)\sigma(M). \tag{9}$$

Here $\overline{E}_M = |M|^{-1} \sum_{x \in M} E(x)$ is mean evidence, and $Q(M) \geq 0$ is the resulting label-free quality score. The implementation uses safeguards of 10^{-6} in the squared-perimeter denominator and 10^{-8} in the hull-area denominator; these do not alter the definitions for nondegenerate candidates. Candidates from both intensity windows are considered jointly, and the delivered segmentation is

$$\widehat{M} = \arg \max_{M \in C_{\text{primary}}} Q(M)\pi(M)^{\gamma}, \qquad \gamma = 1. \tag{10}$$

In Eq. (10), C_{primary} is the union of all recorded candidates from both windows, $\widehat{M}$ is the delivered mask, and γ is the dimensionless persistence exponent. The factor $\pi(M)^{\gamma}$ controls how strongly threshold stability modifies the region-quality ranking: $\gamma = 0$ ignores persistence, whereas increasing γ increasingly favours components that survive more sampled threshold levels without exceeding the 6% plateau-growth tolerance. Its intended role is to down-rank transient high-Q components produced by a narrow threshold range while retaining candidates supported across the sweep. The tested values $\gamma \in \{0, 0.5, 1, 2\}$ are compared on TCGA-LGG; $\gamma = 1$ gives a linear persistence weight and is then frozen for external evaluation. Persistence is therefore a stability modifier, not evidence of tumour identity, and cannot by itself establish tumour specificity.

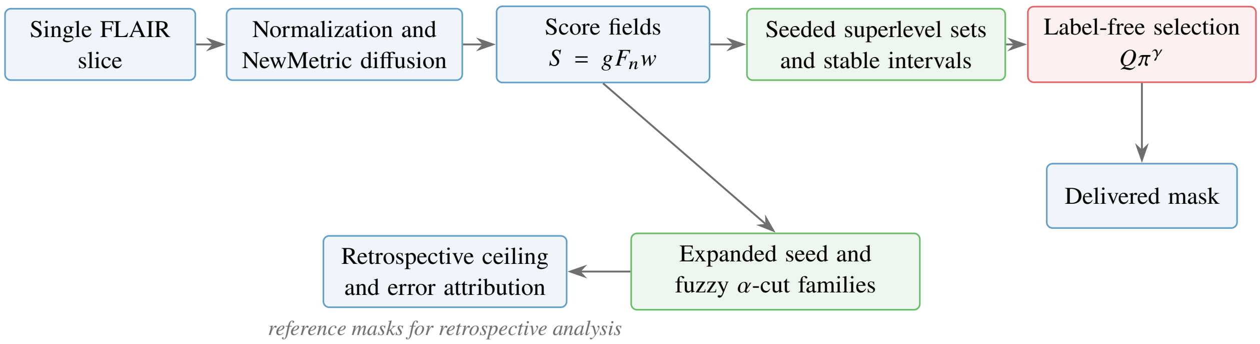

Figure 1: **Method and analytical roles.** The upper path produces the label-free mask. Expanded seed and fuzzy α-cut families quantify candidate capacity and support retrospective error attribution; they do not alter the primary output. Reference masks are used only in the explicitly marked development and retrospective analyses.

3.5 Conditional Brain-ROI Renormalization

In skull-intact TCGA images, bright extracranial structures can determine the normalization maximum and compress intra-brain tumour contrast. Let $R \subseteq \Omega$ denote the final morphological brain ROI. To construct it, let $T_O(Z)$ denote Otsu's threshold computed from the values in a finite set Z. The threshold $0.5T_O(\{I(x) : I(x) > 0\})$ first defines a head foreground; its largest 8-connected component is hole-filled. A 14-pixel disk erosion defines an inner core and hence a peripheral band. Pixels in this band above T_O of the head-foreground intensities are removed as bright shell. The remainder undergoes a one-pixel disk opening, removal of components smaller than 64 pixels, retention of the largest component, hole filling, and a two-pixel disk closing; call this preliminary brain set R_0. A component C is treated as a dark orbital candidate only if it is an 8-connected component of $\{x \in R_0 : I(x) < P_{25}(I; R_0)\}$, has area $60 \leq |C| \leq 0.05|R_0|$, has circularity $4\pi|C|/\text{per}(C)^2 > 0.45$, and has its centroid within the anterior 42% of the ROI height, and lateral displacement greater than 10% of the R_0 width from the midline. The detected component, its midline reflection, and their 13-pixel disk dilations are removed. The resulting binary set is the final ROI R; if no component satisfies the orbital rules, $R = R_0$. The score fields and primary candidate family are then reconstructed within this ROI. Let M_0 and M_1 denote the masks selected before and after ROI renormalization, respectively, and define the ROI containment fraction of a nonempty mask M as $f_{\text{in}}(M) = |M \cap R|/|M| \in [0, 1]$. The ROI-renormalized mask is adopted only if

$$f_{\text{in}}(M_0) < 0.65 \quad \text{and} \quad f_{\text{in}}(M_1) \geq 0.50; \tag{11}$$

otherwise M_0 is retained. The condition is inactive for already skull-stripped images and is therefore not independently validated by the two external cohorts.

The two adoption thresholds are developmental choices. Their sensitivity is audited by varying the first-mask threshold over 0.55–0.75 and the ROI-candidate threshold over 0.40–0.60. Tumour size and peripheral location are examined retrospectively using the upper quartiles of reference-mask area and normalized reference-centroid eccentricity; these quantities are not inputs to the gate.

3.6 Expanded Candidate-Capacity Analysis

The expanded analysis tests whether a suitable boundary is absent from the candidate family or present but not recoverable by label-free ranking. In addition to seeded superlevel-set candidates, four-centre one-dimensional fuzzy c-means [13] is applied to the values $\{I_n(x) : x \in B\}$, producing memberships $\mu_k(x) \in [0, 1]$ for intensity class $k \in \{1, 2, 3, 4\}$. The darkest class is excluded, and 4-connected components of the binary membership sets $\{x \in B : \mu_k(x) \geq \alpha\}$ for the remaining classes are added as seed-independent boundary hypotheses. These candidates are used for capacity measurement, not for the primary delivered segmentation.

Let N be the number of cases in the cohort, $i \in \{1, \ldots, N\}$ index a case, and $Y_i \subseteq \Omega$ be its binary whole-tumour reference mask. Nested seed budgets $s \in \{10, 15, 20, 25, 30\}$ per window and nested counts $a \in \{0, 6, 11, 15, 23, 39\}$ of membership levels are evaluated without removing candidates admitted at smaller budgets. The six-level grid is $\{0.35, 0.45, 0.55, 0.65, 0.75, 0.85\}$; successive grids are its union with 5, 9, 17, and 33 equally spaced levels on $[0.25, 0.95]$, respectively. Let $C_{i,s,a}$ denote the resulting nested candidate set for case i. Then

$$U_i(s, a) = \max_{M \in C_{i,s,a}} D(M, Y_i), \qquad U(s, a) = \frac{1}{N} \sum_{i=1}^{N} U_i(s, a), \tag{12}$$

$U_i(s, a) \in [0, 1]$ is the case-level pool oracle and $U(s, a)$ is its cohort mean; both are retrospective candidate-family ceilings. Practical saturation requires the upper bounds of the paired bootstrap 95% confidence intervals for both final budget increments to be smaller than the prespecified margin $\varepsilon_{\text{sat}} = 0.005$ Dice.

Computational cost is measured over all 110 TCGA slices on an Intel Core Ultra 7 265K CPU (20 physical cores, 31.6 GiB RAM, Windows 11). The full canonical timing includes NewMetric preprocessing, candidate generation and label-free selection. The expanded-budget timing is recorded separately and measures preprocessing plus the additional candidates through 30 seeds per window and 39 nested α levels; it loads the frozen canonical pool and is therefore not a second end-to-end inference time. The worker count used by the historical canonical batch run is not stored in its provenance, whereas the expanded-budget audit runs serially; the reported values are therefore descriptive computational costs rather than standardized serial-latency benchmarks.

3.7 Operational Error Decomposition and Evaluation Measures

Aggregate overlap does not uniquely identify why a segmentation system fails: region-, boundary-, and distance-based measures respond to different error properties, and no single measure captures all of them [34, 35]. More generally, error-analysis frameworks for detection and instance segmentation separate localization and missed-target errors and use oracle interventions to expose performance hidden by an aggregate score [36]. Candidate-based segmentation creates an additional distinction: a suitable mask may be absent from the generated pool or present but ranked below another candidate. The decomposition below adapts these principles to the generate–select structure of this study; it is not adopted from an established clinical taxonomy.

For a given case, let $Y \subseteq \Omega$ denote its nonempty whole-tumour reference mask, let M_j be candidate j, and let $\widehat{M}$ be the delivered mask from Eq. (10). Candidate localization is summarized by $R_{\max} = \max_j |M_j \cap Y|/|Y| \in [0, 1]$, the greatest fraction of the reference tumour reached by any single candidate. Here "localization" means candidate reach or target coverage within the generated pool; it does not mean automatic slice detection, centroid localization, or clinical lesion detection. Boundary availability is summarized by the pool-oracle Dice $D_{\max} = \max_j D(M_j, Y) \in [0, 1]$, using the Dice definition given above. The selection gap is $\Delta_{\text{sel}} = D_{\max} - D(\widehat{M}, Y) \geq 0$; it measures the Dice loss associated with label-free ranking relative to the best mask already available in the same pool.

Cases are assigned to mutually exclusive diagnostic states in the order in which a generate–select system can first become limiting:

(i) $R_{\max} < 0.50$: *candidate-reach limitation* (localization failure), because no generated candidate covers a majority of the reference target;

(ii) $R_{\max} \geq 0.50$ and $D_{\max} < 0.70$: *boundary-representation limitation*, because the pool reaches most of the target but contains no candidate meeting the operational overlap level;

(iii) $D_{\max} \geq 0.70$ and $\Delta_{\text{sel}} \geq 0.10$: *selection loss*, because a representable mask exists but the delivered mask loses at least 0.10 Dice relative to it;

(iv) all remaining cases: *no classified major bottleneck* (reported compactly as near-ceiling/successful).

The hierarchy identifies the earliest *observable* limitation in the candidate-generation–selection sequence. It does not prove a unique causal or biological failure mechanism: a case assigned to an earlier state may also contain later-stage weaknesses. Nor are the cut-offs validated clinical acceptability limits. The 0.50 target-coverage cut-off operationalizes majority reach, 0.70 Dice operationalizes a usable candidate for this decomposition, and a 0.10 Dice gap operationalizes a material within-pool selection loss. They are study-specific analysis conventions and are not used to generate or select a mask.

Descriptive sensitivity is assessed over the complete $3 \times 3 \times 3$ grid $R_{\max} \in \{0.40, 0.50, 0.60\}$, $D_{\max} \in \{0.60, 0.70, 0.80\}$, and selection-gap threshold $\in \{0.05, 0.10, 0.15\}$. This audit changes only the retrospective class labels, not the predicted masks. Its purpose is to distinguish a conclusion that persists over reasonable operational definitions from a case count that depends on one selected threshold triplet. As an independent check on the interpretation of $R_{\max}$, the analysis also reports whether at least one automatic seed lies inside Y; seed hit is not part of the class definition.

The primary performance measure is Dice, calculated over all cases including complete misses. For prediction A and reference Y, Jaccard overlap is $J(A, Y) = |A \cap Y|/|A \cup Y|$, sensitivity is $|A \cap Y|/|Y|$, and precision is $|A \cap Y|/|A|$ (defined as zero for an empty prediction). The number of zero-Dice cases and the numbers exceeding Dice thresholds of 0.50 and 0.70 are secondary measures. Boundary agreement is characterized by average symmetric surface distance (the mean of the bidirectional nearest-boundary-pixel distances), 95th-percentile Hausdorff distance (the 95th percentile of their pooled bidirectional distances), and boundary F1 within two pixels. Boundary precision and recall are the fractions of predicted and reference boundary pixels, respectively, lying within Euclidean distance two pixels of the opposite boundary; their harmonic mean is boundary F1. All distances are in pixels and are summarized only when the two masks overlap [34]. Lesion-level sensitivity, precision and F1 use 4-connected predicted and reference components, with a pair counted as matched when its intersection over union, $\text{IoU} = |A \cap Y|/|A \cup Y|$, is at least 0.10.

Boundary strength is measured on the one-pixel inner boundary $Y \setminus \text{erode}(Y)$ and one-pixel outer boundary $[\text{dilate}(Y) \setminus Y] \cap B$; their union is denoted $\partial_1 Y$. The primary descriptor, the weak-boundary fraction, is $|\{x \in \partial_1 Y : H(x) \leq \text{median}_B H\}|/|\partial_1 Y|$. Additional descriptors include the median absolute derivative of F_n along the signed-distance normal to Y; the fraction of those directional derivatives with the same sign as their median; and the 1-Wasserstein distance between F_n values in three-pixel inner and outer rings, divided by the interquartile range of F_n in B. Tumour area is $|Y|$, and reference compactness is $4\pi|Y|/p(Y)^2$, using the external-contour perimeter $p(\cdot)$ defined above. These quantities use ground truth only for retrospective error attribution.

3.8 Statistical Analysis

The analysis record is one prespecified axial slice per imaging study. TCGA-LGG and UCSF-PDGM contribute one study per patient. The 150 BraTS studies represent 143 unique patients; consequently, the patient, rather than the study, is the independent resampling unit whenever BraTS contributes to an analysis, and repeated studies from the same patient remain in the same resampling cluster. Arithmetic means retain zero-Dice cases. Unless stated otherwise, uncertainty is reported by two-sided percentile-bootstrap 95% confidence intervals (CIs) from 20,000 resamples. A single case is resampled for an unpaired cohort summary, whereas the within-case difference is resampled for a paired comparison.

For paired mean-Dice comparisons, a two-sided Monte Carlo sign-flip test independently reverses the sign of each within-case difference under the null hypothesis; 20,000 sign configurations are sampled, and one is added to both the exceedance count and its denominator. The three pairwise smoothing-ablation tests constitute one family and their p-values are adjusted by the Holm procedure. Parameter sweeps, candidate-budget analyses and all comparisons used to choose TCGA parameters are developmental; their CIs quantify uncertainty within the development set and are not interpreted as independent confirmatory tests.

Spearman's ρ quantifies the association between each boundary descriptor and the candidate ceiling. Boundary-analysis CIs use 20,000 patient-clustered resamples. For the adjusted analysis, the boundary descriptor, candidate ceiling, log-transformed tumour area, compactness and normalized three-pixel ring intensity separation are first converted to ranks. The ranked descriptor and ranked ceiling are separately regressed on the three ranked covariates, and the Pearson correlation between the two residual vectors defines the partial Spearman coefficient. Its CI uses 5,000 patient-clustered resamples. The candidate-saturation CIs use paired case resampling. All bootstrap and Monte Carlo procedures use fixed, analysis-specific random seeds to permit exact computational reproduction; their values are recorded in the analysis scripts and provenance files. The UCSF grade comparison is exploratory and uses a two-sided Mann–Whitney U test, Cliff's δ [37], and an independent 20,000-resample bootstrap CI for the grade-3 minus grade-2 mean difference; no multiplicity adjustment is applied to this single exploratory comparison.

Table 2: Fixed parameters of the primary label-free method and the retrospective capacity analysis.

Stage	Parameter	Fixed value
Normalization	Non-background support	$I > 0.05$
NewMetric	β_{NM}, Δt, iterations	5.0, 0.15, 3
Edge field	Gradient and scale	Sobel magnitude; intra-support P_{95}
Intensity weighting	Lower bound; softness	0.28; 0.05
Intensity weighting	Upper windows	0.82 and 0.97
Seed localization	FLAIR percentile bands	$[30, 88]$ and $[30, 98]$
Seed localization	Maximum per window; spacing	10; 12 pixels
Region evolution	Threshold range; levels	$S(s)$–$0.20S(s)$; 200
Region evolution	Area cap; jump ratio	$0.25\lvert B\rvert$; 4
Region evolution	Acceleration ratio; edge-wall mean	2.5; 0.08
Stable plateau	Relative area tolerance; minimum area	0.06; 30 pixels
Instability stops	Activation area $n_{\min}$	$\max(40, 0.02\lvert B\rvert)$ pixels
Persistence	$\pi(M)$	Number of sampled levels in plateau
Selection	Persistence exponent γ	1
ROI condition	Adoption thresholds	$f_{\mathrm{in}}(M_0) < 0.65$, $f_{\mathrm{in}}(M_1) \geq 0.50$
Capacity analysis	Seed budgets	$10, 15, 20, 25, 30$ per window
Capacity analysis	Nested α counts	$0, 6, 11, 15, 23, 39$
Saturation	Practical Dice margin $\varepsilon_{\mathrm{sat}}$	0.005

4 Results and Discussion

4.1 Primary Segmentation Performance

On the TCGA-LGG development cohort, the primary label-free method achieves a mean Dice of 0.7893 and a median of 0.9134 over all 110 cases; six cases have zero Dice. Dice is at least 0.70 in 88/110 cases. These values describe the two-window, ten-seed-per-window superlevel-set candidate family selected by $Q\pi^{\gamma}$ with $\gamma = 1$; they do not include fuzzy α-cut candidates or ground-truth oracle selection.

The same primary candidate family contained 53,002 masks in total, with a median of 489.5 candidates per case. Retrospective oracle selection raises the mean Dice to 0.8722, leaving a mean oracle–selected difference of 0.0829 (Table 3). The oracle value is an attainable ceiling within the generated family and is not a label-free segmentation result. The coexistence of a high median delivered Dice and a non-negligible mean oracle gap indicates a heterogeneous error distribution: most cases are segmented well, whereas a smaller subset incurs either candidate-generation or candidate-selection failure.

Table 3: Primary TCGA-LGG result and retrospective capacity of the same candidate family. All means include zero-Dice cases.

Measure	Result
Label-free mean / median Dice	0.7893 / 0.9134
Zero-Dice cases	6/110
Cases with Dice ≥ 0.70	88/110
Median candidates per case	489.5
Pool-oracle mean Dice	0.8722
Mean oracle–selected difference	0.0829

4.2 Localization–Boundary–Selection Error Decomposition

The hierarchical analysis localizes the source of error more precisely than the cohort mean (Table 4). Candidate localization succeeds in 106/110 cases, and a candidate with Dice ≥ 0.70 is available in 103/110. Thus, only three of the 106 localized cases are classified as boundary-representation failures. Among the 103 boundary-representable cases, 16 retain an oracle–selected difference of at least 0.10 and are therefore classified as selection failures. The remaining 87 cases are near-ceiling/successful.

The 27-setting threshold audit yields 3–5 localization failures, 2–11 boundary-representation failures, 12–19 selection failures, and 81–87 near-ceiling cases. Canonical counts remain unchanged when the oracle threshold varies from 0.60 to 0.70 and when the selection-gap threshold varies from 0.10 to 0.15; the largest redistribution occurs at the stricter oracle threshold of 0.80. Selection failures outnumbered localization and boundary failures throughout the grid, although the exact counts remain definition-dependent.

Table 4: Hierarchical error decomposition on TCGA-LGG. Reference masks are used only retrospectively for attribution.

Error class	Cases	Selected Dice	Oracle Dice
Localization failure	4	0.084	0.115
Boundary-representation failure	3	0.166	0.437
Selection failure	16	0.394	0.844
Near-ceiling / successful	87	0.916	0.927

These results identify two distinct residual limitations. Candidate generation is insufficient in seven cases: four are not adequately localized and three are localized but lack a boundary candidate meeting the representability criterion. In a larger group, however, an adequate candidate already exists and is not selected. Accordingly, increasing candidate diversity and improving label-free ranking address different scientific problems; a higher candidate ceiling cannot be interpreted as improved delivered segmentation unless the selection rule can recover that additional capacity.

Figure 2 illustrates the operational meaning of the four classes using representative cases chosen near the median selected Dice within each class. Candidate reachability may fail before a plausible tumour boundary is generated; alternatively, an adequate pool-oracle boundary may be present but rejected by the label-free selector. The near-ceiling example shows agreement between the available oracle boundary and the delivered mask.

To balance the failure-oriented audit with the method's typical successful behaviour, Figure 3 shows four cases from the near-ceiling/successful class. The cases are selected reproducibly by dividing that class into quartiles of reference-tumour area and, within each quartile, choosing the case closest to the class median selected Dice (0.932). This rule avoids selecting visually favourable extremes while spanning a fourfold range of lesion area. Only the genuinely label-free selected boundary is shown against the reference; no pool-oracle mask enters the displayed method output.

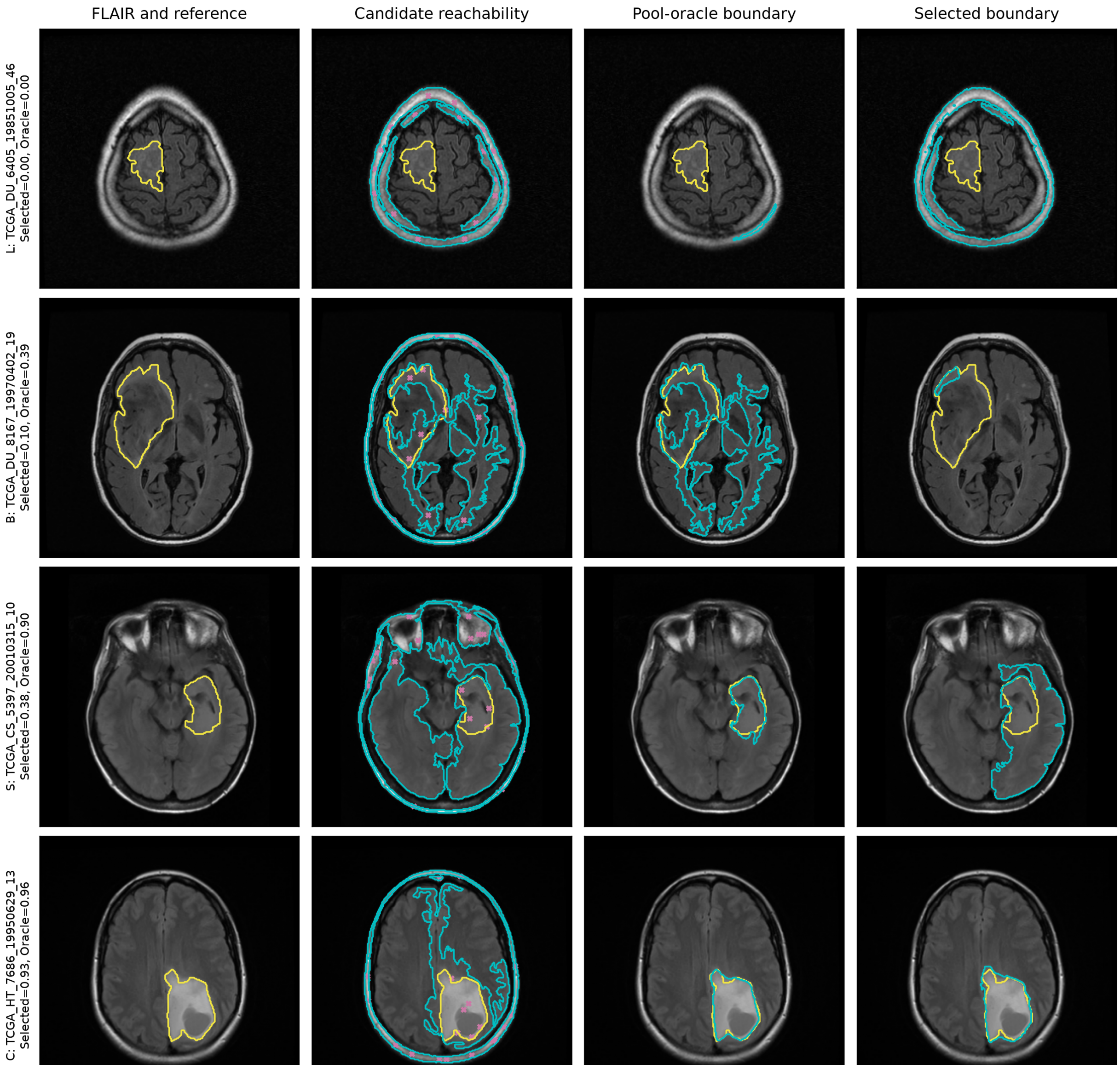

Figure 2: **Operational error-decomposition examples.** Rows represent candidate-reach limitation (L), boundary-representation limitation (B), selection loss (S), and no classified major bottleneck (C). Columns show the FLAIR reference, candidate coverage and seeds, retrospective pool oracle, and label-free output. Yellow is the reference, cyan the indicated boundary, and magenta the seeds; reference-based oracle information is diagnostic only.

4.3 Candidate-Channel Contribution and Practical Saturation

The matched channel analysis confirms this separation (Table 5). The seeded superlevel-set family alone delivers 0.7893 Dice and has an oracle ceiling of 0.8722. The fuzzy α-cut family alone is weaker in delivered performance (0.6473) and oracle capacity (0.8010). Adding it to the seeded family increases the oracle ceiling by 0.0275, from 0.8722 to 0.8997, eliminates cases with zero oracle overlap, and raises Dice-$\geq$ 0.70 candidate coverage from 103/110 to 107/110. In contrast, label-free delivered Dice changes by 0.0000 and remains 0.7893. Thus, the α-cut channel produces a demonstrable gain in candidate availability, but the existing selector does not recover that added capacity as improved delivered segmentation.

Increasing the nested seed and α budgets raises the oracle ceiling from 0.8722 to 0.9055, a paired mean increase of 0.0333 (95% CI [0.0080, 0.0661]), and provides a Dice-$\geq$ 0.70 candidate in 108/110 cases (Table 6). The mean gain from the final seed increment (25 $\rightarrow$ 30) is 0.000025 with an upper 95% confidence bound of 0.000075; the final α increment (23 $\rightarrow$ 39 levels) produces no further mean gain. Both upper bounds are below the prespecified 0.005 Dice margin. The examined family therefore reaches practical saturation near 0.906. This finite experiment does not establish a universal single-FLAIR limit or preclude a different candidate generator

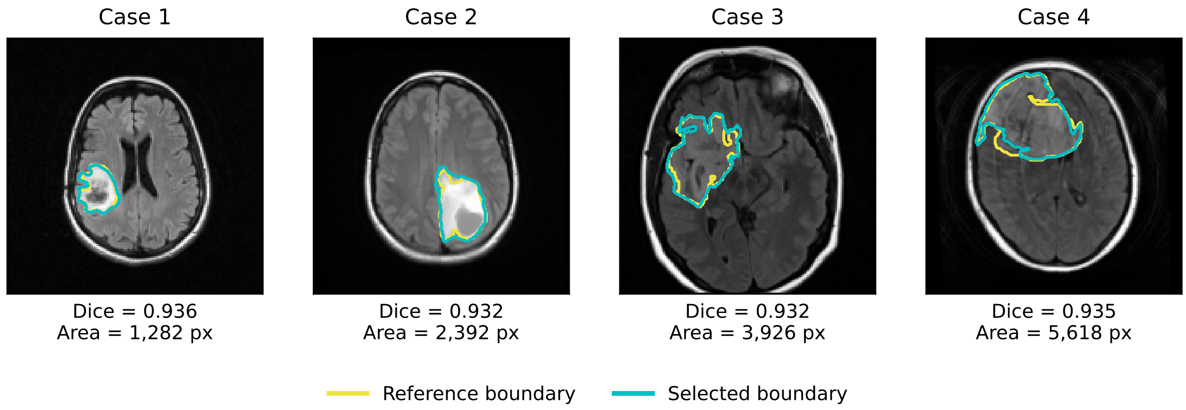

Figure 3: **Successful segmentations across lesion-size strata.** Columns represent successive quartiles of reference-tumour area within the near-ceiling/successful class. Each FLAIR image overlays the retrospective reference boundary (yellow) and the label-free selected boundary (cyan).

Table 5: Matched candidate-channel analysis on TCGA-LGG. Oracle values require the reference mask and are not label-free performance.

Candidate family	Median count	Delivered Dice	Oracle Dice	Oracle zero
Seeded superlevel sets	489.5	0.7893	0.8722	3
Fuzzy α-cuts	159.0	0.6473	0.8010	0
Union	648.5	0.7893 (+0.0000)	0.8997 (+0.0275)	0

from attaining a higher ceiling.

The full canonical method requires a mean of 1.659 s per slice (median 1.688 s; IQR 1.508–1.821 s). The separately timed expanded-budget workload requires a mean of 3.984 s per slice (median 4.059 s; IQR 3.339–4.474 s). The latter is an additional research-analysis cost with a different scope and should not be added to or compared directly with the end-to-end canonical time. Because concurrency was not preserved in the historical canonical provenance, neither value is presented as a deployment latency benchmark.

Table 6: Nested candidate-capacity analysis on TCGA-LGG.

Candidate budget	Oracle Dice	Difference	95% CI
10 seeds, no α-cuts	0.8722	—	—
15 seeds, no α-cuts	0.9010	+0.0288	—
10 seeds, six α levels	0.8997	+0.0275	—
15 seeds, six α levels	0.9048	+0.0326	—
30 seeds, 39 α levels	**0.9055**	**+0.0333**	**[0.0080, 0.0661]**

4.4 Boundary Strength and Attainable Accuracy

The fraction of weak locations along the reference tumour boundary is negatively associated with the candidate-pool ceiling in all three cohorts (Table 7). Spearman correlations are −0.537 in TCGA-LGG, −0.438 in UCSF-PDGM grade 2–3, and −0.467 in BraTS 2023. After adjustment for tumour area, compactness and ring contrast, the partial associations remain negative (−0.325, −0.321, and −0.319, respectively), with confidence intervals excluding zero. Pairwise confidence intervals for cross-cohort differences in the unadjusted correlations include zero.

The replicated direction supports the interpretation that weak, infiltrative FLAIR boundaries constrain the

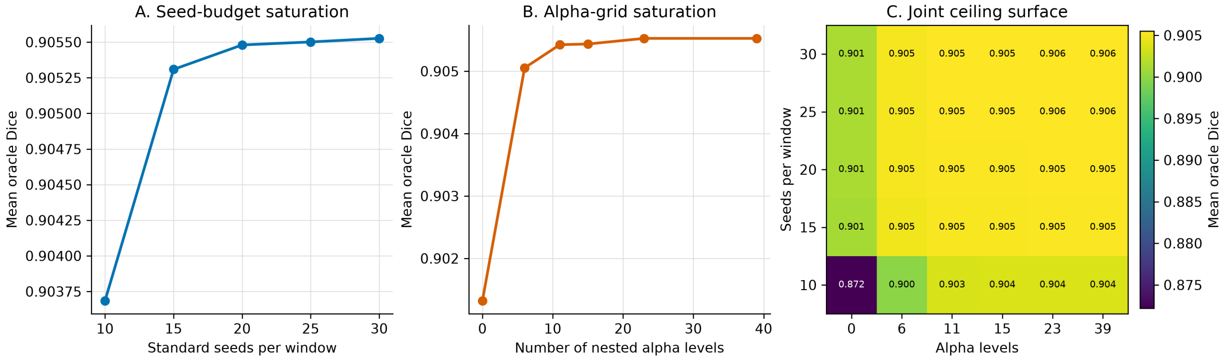

Figure 4: **Practical saturation of candidate capacity.** Nested increases in seed and fuzzy α-cut budgets approach a stable retrospective pool-oracle mean Dice. Oracle values are not label-free performance.

Table 7: Association between weak-boundary fraction and pool-oracle Dice.

Cohort	N	Spearman ρ	Bootstrap 95% CI
TCGA-LGG	110	−0.537	[−0.668, −0.379]
UCSF-PDGM grade 2–3	99	−0.438	[−0.586, −0.259]
BraTS 2023 GLI	150	−0.467	[−0.593, −0.321]

ability of this candidate family to represent an accurate contour. The association persists beyond measured size, shape and local contrast differences, but remains observational and does not demonstrate that weak gradients causally determine segmentation failure.

4.5 External Validation and Cross-Grade Stress Test

Without parameter adjustment, the primary method achieves a mean Dice of 0.7124 (95% CI $[0.6491, 0.7724]$), median Dice of 0.8662, and 11 zero-Dice cases in the 99-case UCSF-PDGM grade 2–3 cohort (Table 8). Mean Dice is 0.6433 for grade 2 and 0.8025 for grade 3. The exploratory grade 3–grade 2 difference is +0.159 (95% CI $[0.044, 0.272]$; Mann–Whitney $U = 1542$, $p = 0.017$; Cliff's $\delta = 0.281$). Because subgroup comparison was not a prespecified confirmatory endpoint, it is interpreted as evidence of greater difficulty in the lower-contrast grade 2 subgroup rather than as a definitive grade effect.

In the BraTS 2023 cross-grade stress test, the same fixed method achieves mean Dice 0.8227, median Dice 0.8927, and one zero-Dice case among 150 studies. The higher BraTS value should not be read as a cross-dataset ranking: BraTS is predominantly high-grade, skull-stripped and template-aligned, whereas TCGA and UCSF serve different clinical and preprocessing regimes. Its role is to show that the fixed method remains operable under a substantial domain and tumour-grade shift.

Table 8: Primary performance by prespecified cohort role.

Cohort and role	N	Mean Dice	Median Dice	Zero Dice
TCGA-LGG, development	110	0.7893	0.9134	6
UCSF-PDGM grade 2–3, external validation	99	0.7124	0.8662	11
BraTS 2023 GLI, cross-grade stress test	150	0.8227	0.8927	1

4.6 Selector Robustness and Conditional ROI Analysis

The persistence exponent shows local tolerance but not unrestricted robustness. Changing γ from 1 to 0.5 yields mean Dice 0.7909, a paired difference of +0.0017 (95% CI $[-0.0180, 0.0230]$). Increasing it to 2 reduces mean

Dice to 0.7477 (difference −0.0416, 95% CI [−0.0796, −0.0084]). Reducing the seed budget from ten to five per window lowers Dice to 0.7087 (difference −0.0806, 95% CI [−0.1277, −0.0396]), while retaining only the narrow window lowers it to 0.7634 (difference −0.0259, 95% CI [−0.0545, −0.0039]). These analyses indicate that performance is relatively stable near the chosen persistence exponent, whereas candidate diversity and the two-window design are consequential.

The conditional brain-ROI renormalization activates in five skull-intact TCGA cases. As a secondary development analysis, it increases mean Dice from 0.7893 to 0.8180 and reduces zero-Dice cases from six to two. The paired mean difference is +0.0288 (95% CI [0.0013, 0.0631]); four cases improve, one deteriorates and 105 remain unchanged. Because both the correction and its adoption condition are developed on TCGA, this value is not the primary result and is not independent validation. The condition never activates in UCSF-PDGM or BraTS 2023, both of which are skull-stripped.

In the threshold audit, changing the ROI-candidate requirement from 0.40 to 0.60 does not alter adoption because every eligible M_1 has $f_{\mathrm{in}}(M_1) = 1$. Raising the first-mask threshold from 0.55–0.60 to 0.65, 0.70, and 0.75 increases adoptions from 4 to 5, 7, and 9, while deteriorations increase from 0 to 1, 3, and 4. The frozen 0.65 value is thus a developmental operating point rather than a theoretically unique constant. No upper-quartile large tumour triggers the frozen gate; one upper-quartile peripheral tumour triggers it and improves. These counts are too small to establish anatomical safety, and the single deterioration among the five frozen adoptions is substantial (Dice change −0.312).

4.7 Matched Training-Free Comparisons

Under the identical TCGA cases, selected slices, FLAIR input and whole-tumour target, Otsu thresholding achieves mean Dice 0.2429, whereas the AUCseg-BT reimplementation achieves 0.3315 (Table 9). The primary method exceeds Otsu by 0.5464 Dice (95% CI [0.4936, 0.5964]) and is better in 103 cases. Its paired difference from the AUCseg-BT reimplementation is 0.4578 (95% CI [0.3844, 0.5297]); it is better in 89 cases and ties in five.

Table 9: Matched training-free comparisons on TCGA-LGG. AUCseg-BT denotes the whole-tumour module reimplemented for the present single-FLAIR 2D protocol, not the complete original multiparametric 3D system.

Method	Mean Dice	Median Dice	Zero Dice	Dice ≥ 0.70
Otsu	0.2429	0.2303	3	0
AUCseg-BT reimplementation	0.3315	0.1551	40	34
Primary method	0.7893	0.9134	6	88

These are matched-protocol controls, not literature-wide superiority claims. They isolate the value of diversified score-topology candidates and label-free selection relative to simpler training-free intensity procedures under the same experimental unit. Comparisons with methods using additional modalities, volumetric context, supervision or prompts remain descriptive rather than rank-based.

4.8 Principal Interpretation

Taken together, the results support error decomposition rather than a single accuracy narrative. The primary method localizes the lesion in 106 cases and represents a Dice-≥ 0.70 boundary in 103, yet fails to recover an available candidate by at least 0.10 Dice in 16 cases. Enlarging the candidate family increases its practically saturated ceiling to approximately 0.906 but does not increase label-free delivery. Candidate generation is therefore not exhausted in the few localization- and boundary-limited cases, while selection is the more prominent remediable bottleneck among representable cases. A future selector trained with labels could potentially exploit this capacity, but it would define a supervised or hybrid method and should not be described as training-free.

5 Limitations

Several limitations define the scope in which these findings should be interpreted. First, TCGA-LGG serves simultaneously as the development and internal evaluation cohort. Reference masks are used to select the largest-tumour slice, determine fixed parameters, quantify performance, assign operational diagnostic states, and calculate candidate-pool ceilings. Although candidate generation and final selection are label-free for each fixed slice, the reported TCGA performance is not an unbiased estimate from a held-out test cohort. The UCSF-PDGM grade 2–3 analysis provides independent institutional validation of the fixed primary method, but it does not remove the risk of development-set optimism in individual design choices. Parameter comparisons, including the persistence exponent, window configuration, seed budget and ROI adoption condition, should therefore be regarded as developmental analyses rather than confirmatory evidence of optimality. The ROI threshold audit further shows that the number of deteriorations increases as the first-mask adoption condition is relaxed, and one of the five frozen-gate adoptions lost 0.312 Dice. The absence of a deterioration in the very small retrospective large/peripheral strata is insufficient to establish anatomical safety.

Second, the experimental unit is a single axial slice selected using the reference tumour mask. The study consequently evaluates segmentation conditional on a known tumour-bearing slice and does not address automatic lesion detection, slice selection, multifocal sampling across a volume, or three-dimensional boundary consistency. Selecting the largest-tumour slice also favours sections with greater lesion representation and may underestimate the difficulty of small or weakly expressed tumour sections. Extension to volumetric FLAIR should be evaluated with patient-level separation, automatic study-level localization and three-dimensional surface measures before clinical or workflow-level claims are made.

Third, the method uses only FLAIR intensity and image-derived geometry. This restricted input is central to the intended low-resource setting, but it excludes complementary information from T1-weighted, contrast-enhanced T1-weighted and T2-weighted MRI. Weak or infiltrative boundaries may therefore remain intrinsically ambiguous in the available observation. The practically saturated oracle Dice of approximately 0.906 is an upper bound only for the finite candidate family examined here; it is neither automatic segmentation performance nor a universal single-FLAIR ceiling. A different handcrafted, learned or multimodal generator could exceed it. Conversely, the gap between this ceiling and the delivered mask cannot be assumed recoverable without labels: the present quality criterion did not convert the additional fuzzy α-cut capacity into higher label-free performance.

Fourth, external evaluation spans only one independent lower-grade cohort. UCSF-PDGM and BraTS 2023 are skull-stripped, whereas the TCGA derivative is skull-intact and differently preprocessed. The BraTS result is therefore a cross-grade and preprocessing-domain stress test, not lower-grade external validation, and cohort means should not be interpreted as direct rankings of tumour grades or datasets. The conditional brain-ROI correction is developed on TCGA and is inactive in both external cohorts; its apparent reduction of complete failures requires confirmation in an independent skull-intact lower-grade cohort. Differences in annotation conventions, acquisition protocols and spatial resolution may also affect overlap and boundary measures. Surface distances reported in pixels are not interchangeable with physical millimetres unless image spacing is harmonized.

Finally, the operational error decomposition, boundary analyses and qualitative case panels are retrospective. The successful examples are selected using reference-tumour area strata and retrospective Dice proximity to the class median; they illustrate typical within-class behaviour rather than prospective case selection or an independent performance estimate. The thresholds defining candidate-reach limitation, boundary-representation limitation and selection loss are study-specific research conventions rather than established or clinical acceptance criteria. The hierarchy identifies the earliest observable bottleneck and does not prove that later weaknesses or other causal mechanisms are absent. Although the 27-setting sensitivity grid preserves the qualitative predominance of selection loss, the exact state counts change with these definitions, especially when the oracle threshold is raised to 0.80. The negative weak-boundary association replicated in direction across all three cohorts, and none of the pairwise confidence intervals for differences in Spearman correlation excludes zero; this supports directional consistency but does not establish equal effect magnitudes. Because the analysis is observational and derives boundary descriptors from the reference contour, it does not prove that weak gradients causally produce low candidate ceilings. Larger prospective multi-institutional cohorts, prespecified primary endpoints and externally fixed decision rules are required to test causal or clinical claims.

6 Conclusions

This study demonstrates that performance in training-free single-FLAIR lower-grade glioma segmentation is more informative when decomposed into localization, boundary representation and label-free selection. On the TCGA-LGG development cohort, the primary method achieves a mean Dice of 0.789, localizes the tumour in 106/110 cases and contains a Dice-$\geq$ 0.70 candidate in 103/110. Among these 103 representable cases, however, 16 retain an oracle–selected difference of at least 0.10. The resulting distinction is fundamental: failure to generate an adequate boundary and failure to select an already available boundary are separate error mechanisms and require different remedies.

The candidate analyses reinforce this conclusion. The primary seeded superlevel-set family has an oracle Dice of 0.872, while nested seed and fuzzy α-cut expansion reaches a practically saturated ceiling of 0.906. Despite this increase in candidate capacity, label-free delivered performance did not improve. More extensive candidate generation is therefore useful for the remaining localization- and representation-limited cases, but it does not resolve the ranking problem in representable cases. Consistently, a larger weak-boundary fraction is associated with a lower attainable ceiling in TCGA-LGG, UCSF-PDGM and BraTS 2023, supporting boundary salience as a reproducible correlate of representability without implying causation.

With all parameters fixed after TCGA development, the method achieves mean Dice 0.712 on the independent UCSF-PDGM grade 2–3 cohort and 0.823 in the BraTS 2023 cross-grade stress test. These results support transfer of the fixed-slice method across institutions and preprocessing regimes, but not direct cross-cohort ranking or volumetric clinical performance. The secondary brain-ROI correction reduces TCGA complete failures, yet remains to be validated in an independent skull-intact lower-grade cohort. Likewise, the present evidence does not isolate an accuracy advantage of NewMetric over alternative edge-preserving operators; its role here is to provide the fixed geometric field within which candidate generation and selection are studied.

The principal contribution is therefore both methodological and diagnostic: a training-free segmentation result should report not only its delivered overlap, but also whether the lesion is localized, whether an adequate boundary exists within the candidate family, and whether the label-free criterion recovers it. For the present single-FLAIR setting, the most immediate opportunity lies in improving candidate selection while preserving the distinction between training-free and label-trained decision rules. Future work should evaluate that decision problem under prespecified patient-level splits, extend the method to automatic three-dimensional analysis, and validate all fixed rules in additional independent lower-grade cohorts.

Author Contributions

S.C.: conceptualization, theoretical formulation, methodology, formal analysis, investigation, validation, and supervision. H.K.: software, implementation, visualization, and writing of the original manuscript. Both authors reviewed and approved the final manuscript.

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Conflict of Interest

The authors declare no competing interests.

Ethics Statement

This study used only de-identified retrospective imaging datasets obtained through their applicable distribution channels and involved no new participant recruitment, intervention, or collection of identifiable information. Ethical approval and informed-consent procedures for the source cohorts were governed by the institutions that originally collected and released those datasets. The present analysis introduced no new contact with participants and used the datasets under their applicable access and data-use conditions.

Data Availability

TCGA-LGG, BraTS 2023 GLI, and UCSF-PDGM are available through their respective controlled or public distribution channels under the terms specified by the data providers. The present study does not redistribute the source MR images or reference masks. Derived case-level measurements and the fixed experimental specification are archived with the accompanying research materials.

Code Availability

The fixed method, data-preparation procedures, evaluation routines, hyperparameters, and manuscript sources are archived in a private GitHub repository at `https://github.com/salim-ceyhan/flair-lgg-error-decomposition/tree/manuscript-release`. Repository access can be provided confidentially to editors and reviewers on request. The repository will be made publicly accessible upon acceptance of the article; source imaging data are not included.

A Geometric Role of the Edge-Preserving Field

The geometric component of the method is confined to construction of the edge-preserving field F_n used by the score in Eq. (6). Following the spatial-feature-manifold formulation [38], a scalar image may be represented as the graph

$$X(x^1, x^2) = \big(x^1, x^2, I(x^1, x^2)\big), \tag{13}$$

whose induced metric couples spatial displacement to intensity variation. The NewMetric operator used here introduces a direction-dependent deformation of this metric and performs a short diffusion with fixed parameters $(\beta_{\mathrm{NM}}, \Delta t, n) = (5.0, 0.15, 3)$ [5]. The numerical result is then normalized to produce F_n; no Gaussian filtering is applied in the primary method.

This construction should be interpreted narrowly. NewMetric supplies the fixed field from which edge permeability, intensity weighting and candidate boundaries are derived. The study does not identify NewMetric itself as the source of the observed Dice performance, nor does it establish superiority over other edge-preserving diffusion operators. The empirically supported contribution is the decomposition of error within the complete training-free method. Accordingly, "Finsler" describes the geometric origin of the image field rather than a stand-alone accuracy claim.

B Seeded Persistence and Candidate Stability

For a score field $S : \Omega \to [0, 1]$, define its superlevel sets $X_t = \{x \in \Omega : S(x) \geq t\}$. As the threshold decreases, these sets form a nested filtration. Given a seed s, the method follows the connected component $C_t(s) \subseteq X_t$ containing that seed. A candidate is recorded when the component area remains within the relative tolerance specified in Section 3.4; the number of consecutive sampled threshold levels in that plateau is its persistence $\pi(M)$.

This quantity is a seed-conditioned, zero-dimensional stability descriptor. It measures how many of the 200 prespecified sweep levels a connected region remains approximately unchanged before growth or merging, but it does not identify that region as tumour. Because it is a discrete run length on an image-specific score sweep, it is used as a within-image ranking attribute rather than as a physical duration or an absolute quantity comparable across images. Its empirical contribution is evaluated rather than assumed: nearby exponents perform similarly, whereas excessive persistence weighting degrades accuracy.

C Supplementary Verified Analyses

C.1 Selector and Candidate-Diversity Sensitivity

Table 10 summarizes the prespecified perturbations of the frozen TCGA candidate records. The result is tolerant to a moderate change in the persistence exponent, but not to excessive weighting, removal of half of the seed budget, or removal of the broad intensity window. These comparisons are development-set sensitivity analyses and are not independent tests of parameter optimality.

Table 10: Sensitivity of label-free selection on TCGA-LGG. Differences are paired against the primary configuration.

Configuration	Mean Dice	Difference	Bootstrap 95% CI
Primary: $\gamma = 1$, ten seeds, two windows	0.7893	—	—
$\gamma = 0.5$	0.7909	+0.0017	$[-0.0180, 0.0230]$
$\gamma = 2$	0.7477	−0.0416	$[-0.0796, -0.0084]$
Five seeds per window	0.7087	−0.0806	$[-0.1277, -0.0396]$
Narrow window only	0.7634	−0.0259	$[-0.0545, -0.0039]$

C.2 Conditional ROI Renormalization

The ROI correction alters only the skull-intact TCGA cohort (Table 11). It is adopted in five cases, improving four and worsening one; all other cases remain unchanged. Its absence of activation on skull-stripped data demonstrates consistency with its intended trigger, not external validation of the correction itself.

Table 11: Secondary analysis of conditional brain-ROI renormalization.

Cohort	Base Dice	ROI-conditional Dice	Zero Dice	Adopted
TCGA-LGG	0.7893	0.8180	$6 \to 2$	5/110
UCSF-PDGM grade 2–3	0.7124	0.7124	$11 \to 11$	0/99
BraTS 2023 GLI	0.8227	0.8227	$1 \to 1$	0/150

C.3 Interpretation Boundaries

All appendix analyses preserve the distinctions used in the main text. Oracle values require reference masks and quantify only the capacity of a specified candidate family. TCGA comparisons are developmental. UCSF-PDGM is the sole independent lower-grade external-validation cohort, whereas BraTS 2023 is a cross-grade stress test. No appendix result changes the primary label-free Dice of 0.7893 or converts the two-dimensional, reference-selected-slice protocol into a fully automatic volumetric setting.

References

[1] M. Havaei *et al.*, "Brain tumor segmentation with deep neural networks," *Med. Image Anal.*, vol. 35, pp. 18–31, 2017. doi: https://doi.org/10.1016/j.media.2016.05.004.

[2] F. Isensee *et al.*, "nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation," *Nat. Methods*, vol. 18, no. 2, pp. 203–211, 2021. doi: https://doi.org/10.1038/s41592-020-01008-z.

[3] N. Gordillo, E. Montseny, and P. Sobrevilla, "State of the art survey on MRI brain tumor segmentation," *Magn. Reson. Imaging*, vol. 31, no. 8, pp. 1426–1438, 2013. doi: https://doi.org/10.1016/j.mri.2013.05.002.

[4] S. Bakas *et al.*, "Advancing the Cancer Genome Atlas glioma MRI collections with expert segmentation labels and radiomic features," *Sci. Data*, vol. 4, art. 170117, 2017. doi: https://doi.org/10.1038/sdata.2017.117.

[5] H. Kılıç, S. Ceyhan, and O. N. Gerek, "A novel family of edge preserving anisotropic filters," *Digit. Signal Process.*, vol. 128, art. 103623, 2022. doi: https://doi.org/10.1016/j.dsp.2022.103623.

[6] S. Bakas *et al.*, "Identifying the best machine learning algorithms for brain tumor segmentation, progression assessment, and overall survival prediction in the BRATS challenge," *arXiv:1811.02629*, 2019. doi: https://doi.org/10.48550/arXiv.1811.02629.

[7] M. Buda, A. Saha, and M. A. Mazurowski, "Association of genomic subtypes of lower-grade gliomas with shape features automatically extracted by a deep learning algorithm," *Comput. Biol. Med.*, vol. 109, pp. 218–225, 2019. doi: https://doi.org/10.1016/j.compbiomed.2019.05.002.

[8] M. Soltaninejad *et al.*, "Automated brain tumour detection and segmentation using superpixel-based extremely randomized trees in FLAIR MRI," *Int. J. Comput. Assist. Radiol. Surg.*, vol. 12, no. 2, pp. 183–203, 2017. doi: https://doi.org/10.1007/s11548-016-1483-3.

[9] A. Kirillov *et al.*, "Segment Anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 4015–4026. doi: https://doi.org/10.1109/ICCV51070.2023.00371.

[10] J. Ma, Y. He, F. Li, L. Han, C. You, and B. Wang, "Segment anything in medical images," *Nat. Commun.*, vol. 15, art. 654, 2024. doi: https://doi.org/10.1038/s41467-024-44824-z.

[11] Y. M. A. Mohammed *et al.*, "A survey of methods for brain tumor segmentation-based MRI images," *J. Comput. Des. Eng.*, vol. 10, no. 1, pp. 266–293, 2023. doi: https://doi.org/10.1093/jcde/qwac141.

[12] S. Kotte, R. K. Pullakura, and S. K. Injeti, "Optimal multilevel thresholding selection for brain MRI image segmentation based on adaptive wind driven optimization," *Measurement*, vol. 130, pp. 340–361, 2018. doi: https://doi.org/10.1016/j.measurement.2018.08.007.

[13] J. C. Bezdek, *Pattern Recognition with Fuzzy Objective Function Algorithms*. New York: Plenum Press, 1981. doi: https://doi.org/10.1007/978-1-4757-0450-1.

[14] A. Khosravanian *et al.*, "Fast level set method for glioma brain tumor segmentation based on superpixel fuzzy clustering and lattice Boltzmann method," *Comput. Methods Programs Biomed.*, vol. 198, art. 105809, 2021. doi: https://doi.org/10.1016/j.cmpb.2020.105809.

[15] S. M. Darwish *et al.*, "A new medical analytical framework for automated detection of MRI brain tumor using evolutionary quantum inspired level set technique," *Bioengineering*, vol. 10, no. 7, art. 819, 2023. doi: https://doi.org/10.3390/bioengineering10070819.

[16] E. S. Biratu *et al.*, "Enhanced region growing for brain tumor MR image segmentation," *J. Imaging*, vol. 7, no. 2, art. 22, 2021. doi: `https://doi.org/10.3390/jimaging7020022`.

[17] B. Zhao, Y. Ren, Z. Yu, J. Yu, T. Peng, and X.-Y. Zhang, "AUCseg: An automatically unsupervised clustering toolbox for 3D-segmentation of high-grade gliomas in multi-parametric MR images," *Front. Oncol.*, vol. 11, art. 679952, 2021. doi: `https://doi.org/10.3389/fonc.2021.679952`.

[18] A. Senbi, T. Huang, *et al.*, "Towards ground-truth-free evaluation of any segmentation in medical images," *arXiv:2409.14874*, 2024. doi: `https://doi.org/10.48550/arXiv.2409.14874`.

[19] H. Edelsbrunner, D. Letscher, and A. Zomorodian, "Topological persistence and simplification," *Discrete Comput. Geom.*, vol. 28, no. 4, pp. 511–533, 2002. doi: `https://doi.org/10.1007/s00454-002-2885-2`.

[20] D. Cohen-Steiner, H. Edelsbrunner, and J. Harer, "Stability of persistence diagrams," *Discrete Comput. Geom.*, vol. 37, no. 1, pp. 103–120, 2007. doi: `https://doi.org/10.1007/s00454-006-1276-5`.

[21] A. François and R. Tinarrage, "Train-free segmentation in MRI with cubical persistent homology," *J. Math. Imaging Vis.*, vol. 68, art. 20, 2026. doi: `https://doi.org/10.1007/s10851-026-01300-1`.

[22] P. Salembier and M. H. F. Wilkinson, "Connected operators: A review of region-based morphological image processing techniques," *IEEE Signal Process. Mag.*, vol. 26, no. 6, pp. 136–157, 2009. doi: `https://doi.org/10.1109/MSP.2009.934154`.

[23] N. Passat, J. Mendes Forte, and Y. Kenmochi, "Morphological hierarchies: A unifying framework with new trees," *J. Math. Imaging Vis.*, vol. 65, pp. 718–753, 2023. doi: `https://doi.org/10.1007/s10851-023-01154-x`.

[24] N. Passat, B. Naegel, F. Rousseau, M. Koob, and J.-L. Dietemann, "Interactive segmentation based on component-trees," *Pattern Recognit.*, vol. 44, nos. 10–11, pp. 2539–2554, 2011. doi: `https://doi.org/10.1016/j.patcog.2011.03.025`.

[25] J. Matas, O. Chum, M. Urban, and T. Pajdla, "Robust wide-baseline stereo from maximally stable extremal regions," *Image Vis. Comput.*, vol. 22, no. 10, pp. 761–767, 2004. doi: `https://doi.org/10.1016/j.imavis.2004.02.006`.

[26] D. N. Louis *et al.*, "The 2021 WHO classification of tumors of the central nervous system: a summary," *Neuro-Oncology*, vol. 23, no. 8, pp. 1231–1251, 2021. doi: `https://doi.org/10.1093/neuonc/noab106`.

[27] B. H. Menze *et al.*, "The multimodal brain tumor image segmentation benchmark (BRATS)," *IEEE Trans. Med. Imaging*, vol. 34, no. 10, pp. 1993–2024, 2015. doi: `https://doi.org/10.1109/TMI.2014.2377694`.

[28] U. Baid *et al.*, "The RSNA-ASNR-MICCAI BraTS 2021 benchmark," *arXiv:2107.02314*, 2021. doi: `https://doi.org/10.48550/arXiv.2107.02314`.

[29] B. Bonato, L. Nanni, and A. Bertoldo, "Advancing precision: a comprehensive review of MRI segmentation datasets from BraTS challenges (2012–2025)," *Sensors*, vol. 25, no. 6, art. 1838, 2025. doi: `https://doi.org/10.3390/s25061838`.

[30] E. Calabrese *et al.*, "The University of California San Francisco preoperative diffuse glioma MRI dataset," *Radiol. Artif. Intell.*, vol. 4, no. 6, art. e220058, 2022. doi: `https://doi.org/10.1148/ryai.220058`.

[31] E. Calabrese *et al.*, "The University of California San Francisco Preoperative Diffuse Glioma MRI (UCSF-PDGM), Version 5," *The Cancer Imaging Archive*, 2022. doi: `https://doi.org/10.7937/TCIA.BDGF-8V37`.

[32] P. Perona and J. Malik, “Scale-space and edge detection using anisotropic diffusion,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 12, no. 7, pp. 629–639, 1990. doi: https://doi.org/10.1109/34.56205.

[33] J. Weickert, *Anisotropic Diffusion in Image Processing*. Stuttgart, Germany: Teubner, 1998, ISBN 3-519-02606-6.

[34] A. A. Taha and A. Hanbury, “Metrics for evaluating 3D medical image segmentation: analysis, selection, and tool,” *BMC Med. Imaging*, vol. 15, art. 29, 2015. doi: https://doi.org/10.1186/s12880-015-0068-x.

[35] V. Yeghiazaryan and I. Voiculescu, “Family of boundary overlap metrics for the evaluation of medical image segmentation,” *J. Med. Imaging*, vol. 5, no. 1, art. 015006, 2018. doi: https://doi.org/10.1117/1.JMI.5.1.015006.

[36] D. Bolya, S. Foley, J. Hays, and J. Hoffman, “TIDE: A general toolbox for identifying object detection errors,” in *Computer Vision–ECCV 2020*, 2020, pp. 558–573. doi: https://doi.org/10.1007/978-3-030-58580-8_33.

[37] N. Cliff, “Dominance statistics: ordinal analyses to answer ordinal questions,” *Psychol. Bull.*, vol. 114, no. 3, pp. 494–509, 1993. doi: https://doi.org/10.1037/0033-2909.114.3.494.

[38] N. Sochen, R. Kimmel, and R. Malladi, “A general framework for low level vision,” *IEEE Trans. Image Process.*, vol. 7, no. 3, pp. 310–318, 1998. doi: https://doi.org/10.1109/83.661181.